

7(a)(i)	P = 0 and Q = 137	A1
	R = -1 and S = 56	A1
7(a)(ii)	lepton(s)	B1
7(b)(i)	(charge of ddd / Y =) $-\frac{1}{3}(e) - \frac{1}{3}(e) - \frac{1}{3}(e) = -1(e)$	B1
	(charge of $\bar{u}d$ / Z =) $-\frac{1}{3}(e) - \frac{2}{3}(e) = -1(e)$	B1
7(b)(ii)	meson: Z / $\bar{u}d$ because consists of a quark and an antiquark	B1
	baryon: Y / ddd because consists of three quarks	B1